25900043 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-BCA(Honours) REGULAR EXAMINATION, MARCH 2025

SECOND SEMESTER

Core Course (CC) - MG2CCRBCA102 - OPERATING SYSTEMS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Very Short Answer Questions

Answer all questions

Each question carries **2** marks

1. Identify one key function of the Timer in an operating system. [K] / [CO1]
2. What is the purpose of the Process Control Block (PCB)? [U] / [CO2]
3. Explain the necessary conditions required for deadlock to occur. [U] / [CO3]
4. How can a process execute its critical section? [U] / [CO3]
5. What is internal fragmentation? [U] / [CO4]

[2x5 = 10]

Part B

Short Answer Questions

Answer any **5** questions

Each question carries **6** marks

6. Differentiate Microkernels and Module based operating system structures. [U] / [CO1]
7. Explain the functions of communications system call. [U] / [CO1]
8. Explain the different states of a process with a diagram. [U] / [CO2]

9. Define Round Robin (RR) scheduling and explain how it works with time quantum. [K] / [CO2]

10. Describe the three types of buffering in interprocess communication [U] / [CO3]

11. What are Monitors? Explain in detail with its appropriate syntax. [U] / [CO3]

12. What are the different types of frame allocation strategies? Explain briefly [U] / [CO4]

[6x5 = 30]

Part C

Essay Questions

Answer any **2** questions

Each question carries **15** marks

13. Explain process scheduling and scheduling criteria in detail and describe different types of schedulers. [U] / [CO2]

14. Discuss in detail how the Banker's Algorithm helps in avoiding deadlock [U] / [CO3]

15. Compare and contrast paging and segmentation in detail . [An] / [CO4]

[15x2 = 30]